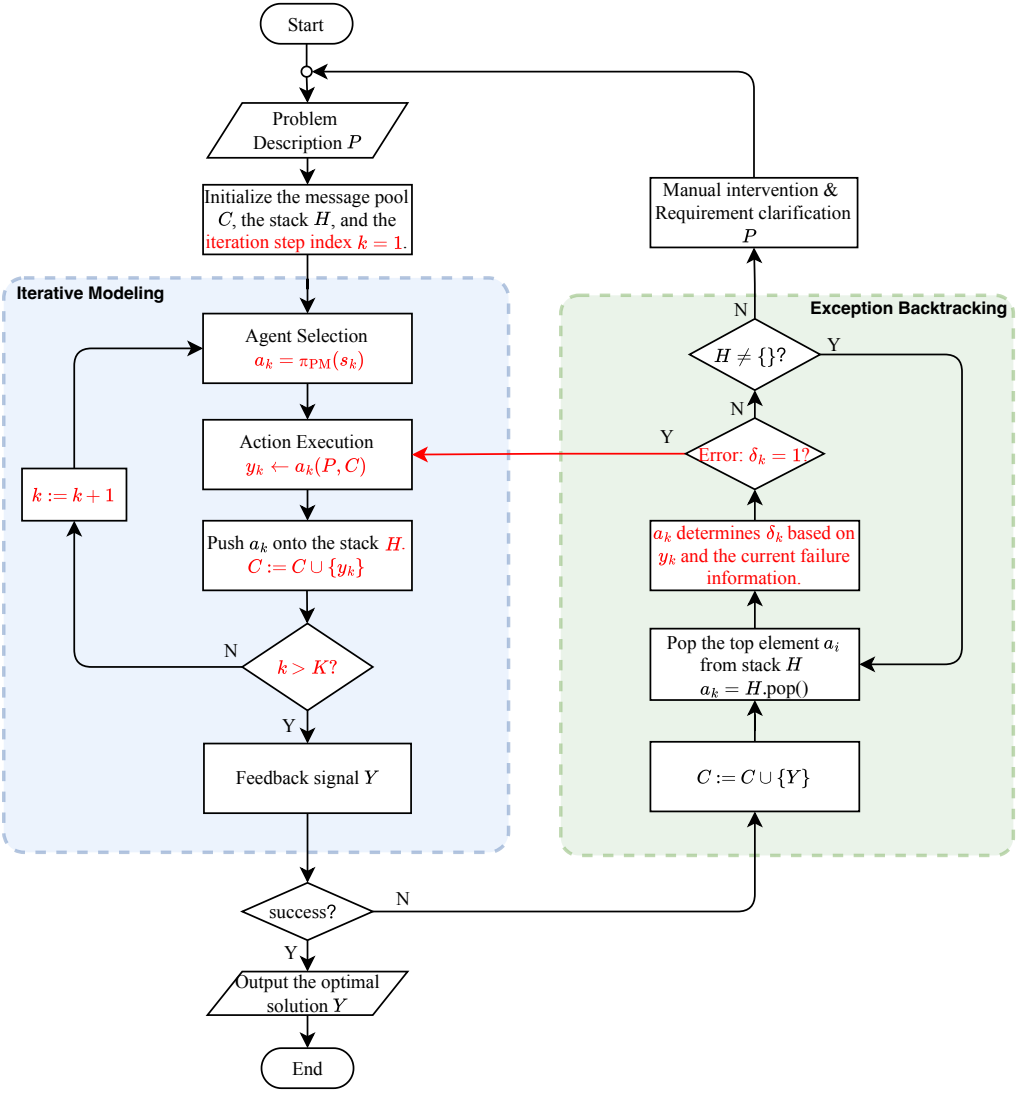
Start
Problem Description P
Initialize the message pool C, the stack H, and the iteration step index $k = 1$.
Iterative Modeling
Agent Selection $a_k = \pi_{\mathrm{PM}}(s_k)$
Action Execution $y_k \leftarrow a_k(P, C)$
Push a_k onto the stack H. $C := C \cup \{y_k\}$
$k > K$?
N
Y
$k := k + 1$
Feedback signal Y
success?
N
Y
Output the optimal solution Y
End
Manual intervention & Requirement clarification P
Exception Backtracking
$H \neq \{\}$?
N
Y
Error: $\delta_k = 1$?
N
Y
a_k determines δ_k based on y_k and the current failure information.
Pop the top element a_i from stack H $a_k = H$.pop()
$C := C \cup \{Y\}$